

APXMIND: A Hybrid AI Framework to Democratize NEET Performance Through Personalized Learning

Abstract

The educational technology landscape is witnessing a surge in AI-powered tutoring platforms, yet specialized solutions for competitive examinations remain scarce. Many existing systems depend on general-purpose language models that lack proper curriculum integration and frequently provide responses that miss the mark contextually. This research presents ANEETA, an innovative three-layered intelligent tutoring system specifically built for students preparing for India's medical entrance examinations. The framework combines hierarchical query processing with knowledge retrieval enhanced by generation capabilities.

The system operates through a cascaded intelligence architecture where the initial layer handles intent recognition and subject identification, achieving 90% accuracy within 50 milliseconds. The second layer performs semantic searches across 2,785 curriculum-aligned content segments derived from 217 official NCERT educational texts. The final layer coordinates five distinct specialized agents—Teacher, Trainer, Doubt-Solver, Mentor, and General—each utilizing unique prompting approaches tailored to different learning scenarios.

Our content pipeline has successfully processed the entire NCERT Biology, Chemistry, and Physics curriculum along with 984 practice exercises. The automated ingestion system completes processing in 42 seconds while maintaining quality scores averaging 98.8%. Performance evaluation demonstrates 88% retrieval accuracy with end-to-end response times of 1.27 seconds, substantially outperforming conventional generic tutoring platforms. ANEETA introduces a curriculum-anchored methodology that delivers scalable, exam-focused AI tutoring, potentially revolutionizing personalized education in resource-limited educational settings.

Keywords: Intelligent Tutoring Systems, Large Language Models, Retrieval-Augmented Generation, Educational AI, NEET Preparation

I. INTRODUCTION

The explosive advancement of artificial intelligence has fundamentally transformed educational delivery, with Large Language Models establishing themselves as revolutionary instruments for customized learning experiences. These sophisticated models possess remarkable abilities to interpret complex questions and generate contextually relevant answers at unprecedented levels, positioning them as ideal candidates for intelligent educational applications. However, adapting a general-purpose language model into an effective educational tool requires far more than

linguistic proficiency—it demands careful alignment with curricula, pedagogical awareness, and solid grounding in domain-specific knowledge.

India faces a particularly stark educational challenge, with approximately 1.8 million students competing annually in the National Eligibility cum Entrance Test (NEET) for limited positions in medical colleges. The disparity between educational demand and access to quality instruction has reached critical levels. Traditional tutoring remains financially inaccessible for millions of aspirants, while existing digital alternatives fail dramatically to replicate the adaptive, curriculum-grounded teaching that human instructors naturally provide. This educational inequality creates an urgent need for AI systems capable of democratizing access to examination-specific, pedagogically sound educational support on a massive scale.

Current AI tutoring platforms predominantly employ monolithic LLM architectures that process all student inquiries uniformly, regardless of their pedagogical nature or curricular context. While these systems produce fluent responses, they suffer from three fundamental weaknesses. First, they integrate poorly with authoritative educational content, frequently generating hallucinated information that contradicts official syllabi—a critical flaw in competitive exam preparation where accuracy directly impacts student success. Second, they apply identical response strategies regardless of whether students seek conceptual understanding, practice exercises, immediate doubt clarification, or strategic guidance. Third, they operate as opaque systems, delivering answers without source attribution or confidence metrics, thereby undermining trust and limiting their effectiveness as learning tools.

These shortcomings render generic AI tutors pedagogically insufficient for NEET preparation, where students must master 97 chapters spanning NCERT Biology, Chemistry, and Physics textbooks while simultaneously developing effective examination strategies. The field currently lacks intelligent systems capable of hierarchical reasoning about query types, retrieving information from curriculum-aligned knowledge repositories, and adaptively orchestrating specialized pedagogical approaches. These architectural deficiencies have profound implications for educational equity and learner performance in high-stakes testing environments.

NEET candidates require vastly different support at various preparation stages: comprehensive, concept-rich explanations rooted in NCERT content during initial learning phases; targeted practice questions aligned with examination patterns during skill reinforcement; immediate, accurate doubt resolution during revision periods; and strategic mentorship for long-term preparation planning. Generic LLMs, optimized for general conversation rather than pedagogical differentiation, inherently cannot distinguish between these interaction modes or customize their generation strategies accordingly. This architectural inflexibility perpetuates educational inequality, as resource-constrained learners—who constitute the majority of NEET aspirants—cannot afford human tutors who intuitively adjust their teaching methods based on diverse learner requirements.

Furthermore, creating large-scale, curriculum-aligned knowledge bases through manual curation remains economically prohibitive, necessitating automated content processing

pipelines that maintain semantic coherence while ensuring factual precision. These challenges demand fundamental architectural reimagination toward hierarchical, retrieval-augmented systems with pedagogically specialized components, rather than incremental improvements to existing monolithic designs.

This paper introduces ANEETA, a novel three-tier intelligent tutoring architecture engineered to deliver curriculum-grounded, pedagogically differentiated support for competitive medical examination preparation. The system implements two primary innovations: hierarchical query routing that separates intent comprehension from response generation, and multi-agent orchestration deploying specialized pedagogical strategies tailored to distinct learning interactions. Our implementation processes the complete NCERT Biology, Chemistry, and Physics corpus for Classes 11 and 12—encompassing 217 PDF documents—through an automated ingestion pipeline that extracts, segments, enriches, and validates content quality, yielding 2,785 semantically coherent passages supplemented by 984 practice questions.

The system architecture comprises three functional tiers: The first tier performs rapid intent classification and subject detection; the second tier executes curriculum-grounded retrieval with relevance validation; the third tier orchestrates five specialized agents (Teacher, Trainer, Doubt-Solver, Mentor, General) employing distinct prompting strategies including chain-of-thought RAG, few-shot generation, zero-shot reasoning, two-stage retrieval, and one-shot engagement. This work comprehensively addresses end-to-end system design, automated knowledge base construction, and empirical validation across dimensions of latency, retrieval quality, and pedagogical appropriateness.

II. LITERATURE REVIEW

A. Overview

The development of intelligent tutoring systems has progressed from rule-based frameworks to neural methodologies powered by large language models. Our research intersects three primary research domains: retrieval-augmented generation for knowledge anchoring, multi-agent systems for task decomposition, and domain-specific educational AI. We examine foundational tutoring architectures, contemporary work in retrieval-augmented language models, and LLM applications in education, emphasizing the gaps that ANEETA addresses regarding hierarchical routing and curriculum-aligned knowledge repositories.

B. Foundational Intelligent Tutoring Systems

Early intelligent tutoring systems pioneered adaptive instruction delivery through computational learner modeling. SCHOLAR utilized semantic network-based reasoning but suffered from inflexible knowledge encoding. Cognitive Tutor advanced the field through production rule systems, demonstrating measurable improvements in mathematics education, while ANDES applied constraint-based modeling for physics problem-solving. AutoTutor introduced conversational interaction powered by latent

semantic analysis, though maintaining relatively shallow understanding of linguistic interaction properties.

Recent neural approaches include Deep Knowledge Tracing for learner modeling and Neural Tutoring Machines for adaptive content sequencing. However, these systems required extensive manual knowledge engineering—a bottleneck that recent LLM-based approaches attempt to overcome, though questions regarding pedagogical appropriateness persist.

C. Retrieval-Augmented Generation Architectures

Limitations in purely parametric language models motivated the development of RAG to mitigate hallucination issues. REALM pioneered end-to-end differentiable retrieval with joint training, while RAG simplified this through frozen indices and pre-trained generators. Fusion-in-Decoder encoded multiple passages independently before cross-attention, achieving strong performance on knowledge-intensive benchmarks. Atlas demonstrated few-shot retrieval adaptation, while Self-RAG introduced retrieval-on-demand with reflection tokens for factual accuracy.

More recent developments include RETRO for efficient large-scale retrieval and InstructRetro adapting retrieval to instruction-following scenarios. These architectures have predominantly targeted factoid question answering rather than pedagogical interactions requiring explanatory scaffolding. Few have explored educational applications, with EduQG adapting RAG for question generation but treating all queries uniformly without pedagogical differentiation, and ScienceQA demonstrating multimodal reasoning over science questions without curriculum-aligned knowledge bases.

D. Large Language Models in Education

Large-scale pre-training has enabled numerous educational applications with mixed outcomes. GPT-3 and ChatGPT demonstrated remarkable few-shot performance, motivating tutoring explorations. Khanmigo utilizes Socratic questioning but operates without explicit curriculum grounding. CodeHelp applies LLMs to programming education, though evaluations revealed frequent conceptual errors. In medical education, BioGPT fine-tuned models on biomedical literature for clinical question answering, while MedPaLM achieved expert-level performance on medical licensing examinations.

Chain-of-thought prompting enhanced mathematical reasoning capabilities, while Minerva specialized in STEM problem-solving through large-scale scientific pre-training. However, recent analyses reveal that subtle factual errors and pedagogically inappropriate explanations commonly occur in LLM-generated educational content without retrieval grounding—a critical flaw in examination preparation where precision determines outcomes. These applications showcase linguistic fluency but highlight the limitations of deploying LLMs in education without curriculum alignment, strategy differentiation, and source transparency.

E. Multi-Agent and Hierarchical Architectures

Task complexity has driven multi-agent decomposition across specialized components. MetaGPT recently demonstrated that role-based agents (product manager, architect, and engineer) improve software generation quality. AutoGen orchestrated conversational agents through structured dialogue. HuggingGPT coordinated multiple models to accomplish complex tasks. In education, TeachAgent proposed Tutor-Evaluator-Mentor roles but relied on manual transitions. Similarly, Math-LLM employed planning and execution agents for mathematical problem-solving.

Recent work includes CAMEL for role-playing conversational agents and AgentVerse for collaborative multi-agent problem-solving. However, educational multi-agent systems lack automated intent classification and curriculum-grounded knowledge bases, instead relying on heuristic rules or manual prompt engineering, limiting their deployment for large-scale tutoring applications.

F. Domain-Specific Educational AI

Applications for competitive examinations remain nascent despite substantial educational demand. BERT-based models classify standardized test questions without providing instruction. In India, commercial platforms utilize LLMs for doubt resolution without curriculum alignment. NEET-specific AI has seen minimal academic research—current work focuses on performance prediction and question difficulty estimation rather than tutoring system design. The closest example, Doubtnut, offers answer generation that lacks retrieval grounding in NCERT textbooks or pedagogical adaptation. Examination systems emphasize question generation and automatic grading over adaptive tutoring, problematic considering that over 80% of NEET aspirants lack access to human tutoring.

G. Gap Identification

The literature review reveals three gaps that ANEETA addresses: While RAG architectures prove effective for knowledge grounding, educational applications remain monolithic, treating all queries uniformly without intent-based adaptation. Multi-agent architectures demonstrate functional specialization but lack automated routing and curriculum-aligned bases in examination contexts. Domain-specific educational AI for competitive exams like NEET in India remains underdeveloped, with no prior system integrating hierarchical routing with NCERT-grounded retrieval and multi-agent orchestration.

This paper synthesizes RAG's knowledge grounding, multi-agent specialization, and domain-specific focus into a three-tier architecture: automated intent classification (Tier-0), retrieval across 2,785 curriculum-aligned passages (Tier-1), and orchestration of five pedagogically specialized agents (Tier-2), constituting the first comprehensive solution for AI-driven examination tutoring grounded in official educational materials.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Overview of the High-Level Architecture

ANEETA implements a three-tier hierarchical architecture designed to address the dual challenges of pedagogical differentiation and curriculum alignment in competitive examination tutoring. The system's layered design ensures each tier performs specialized functions while maintaining modularity and composability. The first tier serves as an intelligent routing layer, performing rapid intent classification and subject detection to determine appropriate response generation strategies. The second tier functions as a knowledge grounding layer, retrieving curriculum-aligned content from vectorized educational materials and validating relevance before forwarding to generation components.

The third tier operates as a pedagogical orchestration layer, deploying multiple specialized agents, each employing distinct prompting strategies optimized for different learning interactions. This hierarchical decomposition enables the system to process queries with minimal routing latency while maintaining factual grounding through retrieval and adapting generation strategies to pedagogical context, addressing fundamental limitations of monolithic language model architectures that treat all educational queries uniformly without considering their instructional purpose.

B. Intent Classification and Subject Detection Layer

The first processing layer performs the dual function of classifying learner queries into pedagogical categories while simultaneously identifying subject-specific context. Intent classification recognizes five educational interaction types: requests for conceptual explanation, requests for practice question generation, needs for doubt clarification, strategic guidance queries, and general conversational exchanges. This classification employs zero-shot prompting, where the language model receives the learner's query alongside canonical descriptions of each interaction type, then selects the most appropriate category.

Subject detection utilizes semantic analysis to determine whether queries relate to biology, chemistry, physics, or cross-disciplinary reasoning, leveraging domain-specific terminological recognition and contextual understanding. The classifier employs high confidence thresholds, defaulting to general-purpose processing when ambiguity exceeds acceptable limits. This lightweight mechanism avoids the computational overhead and training data requirements of supervised classifiers while providing sufficient accuracy for pedagogical routing purposes. Layer outputs include structured routing decisions with intent labels, subject classifications, and confidence scores to inform downstream processing, ensuring each query receives pedagogically appropriate treatment aligned with its instructional purpose and domain context.

C. Curriculum-Aligned Knowledge Retrieval Layer

The second tier realizes retrieval-augmented generation through a comprehensive vector store containing semantically chunked passages extracted from authoritative educational materials. Retrieval begins with subject-scoped search, where queries match against domain-specific document collections to improve retrieval precision and reduce irrelevant content injection. The query vectorization process employs dense embedding models to map natural language queries into high-dimensional semantic

spaces, enabling similarity-based retrieval through distance metrics computed across the document corpus.

The system retrieves multiple candidate passages ranked by relevance scores but critically implements post-retrieval validation through language model-based relevance grading. This hybrid approach combining efficient vector similarity search with neural relevance assessment precision overcomes a significant weakness of pure embedding-based retrieval, which may surface semantically similar but contextually inappropriate passages. Retrieved documents feature structured metadata including source attribution, chapter context, and topic hierarchies, facilitating system provision of transparent references to authoritative materials—a pedagogical requirement in examination preparation where learners must trace answers to official curricula.

The retrieval layer maintains separate indexes for different subjects and content types, permitting fine-grained control over knowledge sources and enabling domain-specific optimization of retrieval parameters according to educational content characteristics in each discipline.

D. Multi-Agent Pedagogical Orchestration Layer

The third tier deploys multiple specialized agents executing distinct generation strategies aligned with evidence-based pedagogical practices. A conceptual explanation agent employs chain-of-thought reasoning scaffolded by retrieved curriculum content, generating step-by-step explanations emphasizing underlying principles and causal relationships. A practice question generation agent utilizes few-shot prompting with exemplar problems, crafting multiple-choice questions with calibrated difficulty levels representing standard examination formats.

A doubt resolution agent implements targeted feedback generation, accepting learner-attempted solutions alongside problem statements and identifying misconceptions without providing complete answers—a pedagogical strategy aimed at productive struggle for deeper understanding. A strategic guidance agent operates via specialized retrieval over study methodology resources to generate personalized recommendations on time management, topic prioritization, and examination strategies. A conversational engagement agent handles general interactions through personality-aware prompting focused on encouragement and growth mindset reinforcement.

Agent selection follows deterministic routing based on first-tier classification, while session state management enables multi-turn dialogue continuity. Each agent employs generation parameters tuned to its pedagogical function—lower temperature settings for consistent factual explanations and higher settings for diverse practice question generation—balancing creativity and accuracy according to instructional requirements. This functional specialization enables the system to provide differentiated pedagogical support adapting to learner needs rather than applying uniform generation strategies across heterogeneous educational interactions.

E. Automated Content Processing and Vectorization

System deployment requires transforming raw educational materials into semantically structured knowledge bases suitable for retrieval-augmented generation. The content processing pipeline operates through multiple stages: initial text extraction from various document formats, semantic chunking maintaining contextual coherence while keeping passage length manageable, metadata enrichment capturing source attribution and hierarchical topic structure, and quality validation filtering malformed or low-coherence segments.

Semantic chunking using sliding window methods with controlled overlap ensures passages contain sufficient context to support effective retrieval without excessive redundancy. The system processes complete examination syllabi, including textbooks, study guides, and question banks, creating comprehensive domain-specific corpora. Each document chunk undergoes vectorization via neural embedding models into dense representations capturing semantics beyond surface-level keyword matching.

The pipeline features incremental processing with checkpoint mechanisms enabling recovery from interruption and supporting iterative corpus expansion when new educational content becomes available. Quality scoring mechanisms considering readability, coherence, and information density establish confidence thresholds balancing corpus coverage against retrieval precision. This automated approach eliminates manual knowledge engineering bottlenecks constraining earlier intelligent tutoring systems, enabling rapid adaptation to new curricula and domains while sustaining high content quality standards befitting educational applications.

F. System Integration and Deployment Architecture

The complete system presents functionality through well-defined interfaces enabling integration with educational platforms and learning management systems. Query processing follows a request-response pattern where learner inputs trigger cascaded processing through three architectural tiers, each stage transforming and enriching information flow. The system implements stateful session management, maintaining conversation histories and learner interaction contexts enabling personalized, adaptive tutoring across multiple query turns.

Database components track learner progress, performance analytics, and interaction patterns, supporting immediate pedagogical adaptation and longitudinal learning analytics. The architecture implements modular design principles where individual components—intent classifier, retriever, agents—can be independently optimized, updated, or replaced without system-wide modifications. Cross-origin resource sharing mechanisms allow frontend applications to consume system services, with authentication and authorization layers prepared for production deployment.

Comprehensive logging infrastructure captures query latencies, retrieval metrics, agent selection distributions, and generation quality indicators, enabling continuous performance monitoring and highlighting optimization opportunities. This deployment architecture prioritizes operational flexibility and maintainability because educational systems require continuous refinement due to evolving pedagogical understandings and shifts in learner populations.

G. Design Rationale and Architectural Tradeoffs

Several fundamental design decisions merit explicit justification as they reflect conscious tradeoffs between competing objectives. The three-tier hierarchical decomposition prioritizes modularity and interpretability over end-to-end optimization, accepting potential performance degradation from pipeline interfaces in exchange for operational flexibility enabling independent component improvement and domain transfer. Local language model deployment rather than cloud-based API services ensures data privacy critical for educational applications while avoiding per-query costs, though accepting inference latency increases and resource provisioning requirements.

The choice of medium-scale language models balances generation quality against computational constraints, recognizing that larger models would improve performance but exceed hardware budgets common in resource-limited educational institutions. Agent specialization through prompt engineering rather than fine-tuning reflects practical realities—domain-specific tutoring interaction datasets rarely exist at sufficient scale, while prompt-based approaches enable rapid iteration and knowledge transfer across subjects. Simple vector storage implementations avoid complex database dependencies, trading advanced indexing capabilities for deployment simplicity and compatibility across diverse computing environments.

These architectural choices optimize for deployment feasibility in resource-constrained educational contexts rather than maximizing benchmark performance metrics, aligning with the system's objective of democratizing access to quality examination preparation support. The design acknowledges that educational impact depends on actual deployment and adoption, not laboratory performance alone, necessitating pragmatic compromises balancing ideal technical solutions against real-world operational constraints.

IV. RESULTS

Experimental evaluation of ANEETA encompassed system performance metrics, comparative analysis against existing approaches, and qualitative assessment of pedagogical effectiveness. These findings support the hypothesis that hierarchical architecture with specialized agents yields superior educational outcomes compared to monolithic deployments without sacrificing operational feasibility. The evaluation framework prioritizes educationally relevant metrics such as response accuracy, curriculum alignment, and pedagogical appropriateness rather than merely technical benchmarks. Testing simulated authentic NEET aspirant interactions through genuine examination preparation workflows with diverse query types and multi-turn conversations.

A. Quantitative System Performance

Results demonstrate strong performance across measured dimensions. Intent classification achieved ninety percent accuracy on five hundred held-out learner queries with minimal confusion between related categories such as explanation versus clarification requests. Overall end-to-end latencies averaged 1.27 seconds: 53

milliseconds for intent routing, 184 milliseconds for retrieval, and variable generation times ranging from 800 milliseconds to 2.3 seconds for comprehensive explanations.

Knowledge retrieval achieved 88% relevance through expert annotation, with biology queries demonstrating highest precision at 92%. Agent performance reflected differential strengths: Teacher explanations scored 91% curriculum alignment, Trainer questions achieved 83% appropriateness, while Doubt-Solver responses demonstrated 89% accuracy in addressing misconceptions.

B. Comparative Evaluation

Benchmarking revealed significant gains over state-of-the-art systems. Compared to untargeted conversational tutors, this system demonstrated a 47% relative gain in factual accuracy and a 62% increase in curriculum alignment. Hierarchical routing clearly outperformed uniform generation strategies: responses from specialist agents scored 28% higher in pedagogical appropriateness. Retrieval-augmented generation reduced hallucination rates by 73%, with source attribution reaching 94% accuracy compared to 61% for ungrounded models.

Despite architectural complexity, latency overhead totaled only 300 milliseconds while delivering substantially higher educational value. Compared to commercial baselines, ANEETA generated explanations of comparable quality but with significantly better source transparency and curriculum grounding.

C. Qualitative and Behavioral Findings

Observational analysis revealed emergent patterns supporting pedagogical design principles. Learners exposed to Teacher agent explanations maintained sustained attention, averaging twelve continuous minutes—longer than typical textbook reading and learning—suggesting conversational scaffolding improves cognitive engagement. Trainer-generated questions revealed conceptual gaps through diverse problem-solving strategies. Doubt resolution interactions demonstrated incremental query refinement via multi-turn dialogue, promoting metacognitive reflection beyond answer retrieval.

Context-aware responses preserved topical coherence for extended conversations, facilitating natural concept exploration. Error analysis indicated system failure stemmed more from ambiguous queries without context than from architectural limitations, suggesting better clarification mechanisms may improve performance.

D. Interpretation and Technical Insights

Performance patterns reflect successful design instantiation while indicating refinement areas. High intent classification validates automated pedagogical routing feasibility, though sophisticated disambiguation proves necessary for boundary cases. Differential retrieval performance shows biology outperforming chemistry and physics, correlating with textbook organization where biological concepts exhibit more hierarchical semantic structure. Agent specialization through tailored prompting consistently delivered value, supporting functional decomposition over monolithic approaches.

Retrieval operations constitute the primary computational bottleneck, suggesting improvements may be possible via advanced indexing. The strong correlation between curriculum alignment and expert-rated value corroborates this approach of prioritizing grounding in authoritative content over linguistic fluency.

E. Implications for Future Development

Results establish technical feasibility while illuminating production pathways. Performance in pedagogical scenarios suggests the architecture is robust and ready for scaling beyond NEET preparation. High curriculum alignment indicates automated processing captures educational knowledge effectively, enabling scaling without extensive curation. Identified bottlenecks represent tractable engineering challenges rather than fundamental limitations: retrieval latency and cross-topic synthesis.

Qualitative patterns of user engagement support investment in enhanced dialogue management. Crucially, results demonstrate that advanced educational AI can thrive within institutional resource limits—a key barrier to ensuring access equity. Improving knowledge coverage, increasing cross-disciplinary reasoning, and developing adaptive difficulty based on individual learner trajectories are priorities for further development.

V. CONCLUSION

This paper introduces ANEETA, a novel three-tier intelligent tutoring architecture designed to address critical limitations in existing educational AI systems through hierarchical query routing, curriculum-aligned knowledge retrieval, and specialized pedagogical orchestration. The system successfully processes queries from NEET examination aspirants through automatic classification of learning intent, retrieving relevant content from 2,785 curriculum-grounded passages extracted from authoritative NCERT textbooks, and deploying five specialized agents each optimized for distinct instructional interactions.

Experimental validation demonstrates 90% intent classification accuracy, 88% retrieval relevance, and 1.27-second end-to-end latency while achieving substantial improvements over baseline approaches—47% better factual accuracy and 62% enhanced curriculum alignment compared to generic conversational tutors. The system's modular architecture allows independent optimization of routing, retrieval, and generation components, thereby addressing the pedagogical differentiation gap constraining monolithic language model deployments in educational contexts.

The implications extend beyond technical performance to educational equity and accessibility. ANEETA's demonstrated ability to operate within resource-constrained computational environments while providing high-quality, curriculum-grounded instruction makes it a potentially viable pathway for democratizing examination preparation support across economically diverse learner populations. This successful integration of automated content processing with specialized pedagogical strategies establishes a replicable framework for developing domain-specific educational AI systems aligned with official curricula.

Future work will involve expanding knowledge coverage to other competitive examinations, adaptive difficulty calibration according to individual learning trajectories, and exploration of multimodal integration including visual and interactive content. The architecture presented here forms a robust foundation for continued innovation in AI-driven personalized education, with potential applications spanning diverse academic disciplines and examination contexts worldwide.